

Customer Churn Prediction Using Machine Learning

1. Introduction

Customer churn prediction is an important business problem that helps companies identify customers who are likely to leave their services. Predicting churn allows businesses to take preventive actions and improve customer retention.

The objective of this project is to build a machine learning model that can predict whether a customer will churn based on demographic and service-related information.

2. Dataset Description

The dataset used in this project is the Telco Customer Churn Dataset.

Dataset Features:

- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure
- Phone Service
- Internet Service
- Contract Type
- Monthly Charges
- Total Charges
- Churn Status

The dataset contains 7043 customer records.

3. Data Preprocessing

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Converted TotalCharges to numeric format.
- Handled missing values using median imputation.
- Converted the target variable (Churn) into binary values.

- Encoded categorical features using Label Encoding.

4. Exploratory Data Analysis

Basic data exploration was performed to understand the dataset structure.

Key observations:

- Most customers did not churn.
- Customers with longer tenure were less likely to churn.
- Contract type had a significant influence on churn behavior.

5. Model Development

The dataset was divided into training and testing sets using an 80:20 split.

Machine Learning Models Used:

1. Logistic Regression
2. Random Forest Classifier

6. Results and Evaluation

Logistic Regression

Accuracy: 81.83%

Classification Metrics:

- Precision (Churn): 0.68
- Recall (Churn): 0.60
- F1 Score (Churn): 0.63

Random Forest

Accuracy: 79.49%

Confusion Matrix

- True Negatives: 931
- False Positives: 105
- False Negatives: 151
- True Positives: 222

Logistic Regression achieved the highest accuracy and was selected as the final model.

7. Business Recommendations

- Focus retention efforts on customers likely to churn.
- Improve customer support for at-risk customers.
- Provide loyalty programs and personalized offers.
- Monitor customer engagement regularly.

8. Conclusion

This project successfully developed a customer churn prediction model using machine learning techniques. Logistic Regression achieved the best performance with an accuracy of 81.83%. The developed model can help organizations improve customer retention and reduce business losses caused by customer churn.